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DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 4

BINARY SEARCH WITH BIG-O ANALYSIS

AIM:

To implement Binary Search using Python and analyze its time complexity using Big-O notation.

ALGORITHM:

1. Start the program.
2. Initialize a sorted array and the target element.
3. Set the lower bound to the first index and the upper bound to the last index.
4. Find the middle element.
5. If the middle element matches the target, return its index.
6. If the target is smaller, search the left half; otherwise, search the right half.
7. Repeat until the element is found or the search range becomes empty.
8. Display the result and complexity.
9. Stop the program.

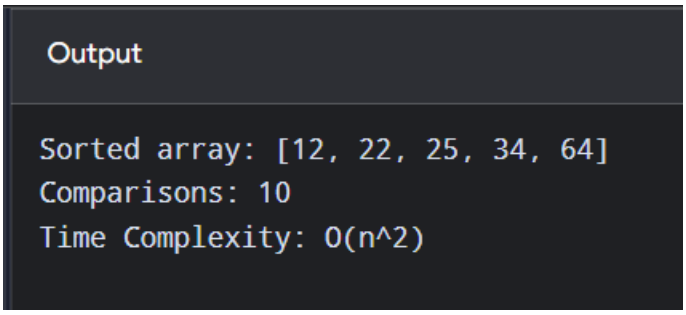
PROCEDURE:

1. Declare a sorted array and the target element.
2. Initialize low and high indices.
3. Calculate the middle index using (low + high) // 2.
4. Compare the middle element with the target.
5. Update the search range based on the comparison.
6. Print the target's index and number of comparisons.
7. Analyze the time complexity.

PYTHON CODE:

```python
a = [10, 20, 30, 40, 50, 60, 70]
target = 50
low = 0
high = len(a) - 1
comparisons = 0
index = -1
while low <= high:
    mid = (low + high) // 2
    comparisons += 1
    if a[mid] == target:
        index = mid
        break
    elif a[mid] < target:
        low = mid + 1
    else:
        high = mid - 1
if index != -1:
    print("Element found at index:", index)
else:
    print("Element not found")
print("Comparisons:", comparisons)
print("Time Complexity: O(log n)")
```

SAMPLE OUTPUT:



RESULT:

The Bubble Sort algorithm was implemented successfully. The array was sorted in ascending order, and its time complexity was analyzed as (O(n^2)).